

AgentLab

A research lab run by agents

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github.com/shuds13/AgentLab

What it is

You give an agent a goal and a system. It runs the investigation itself.

A lab run by agents

A campaign agent submits work to an HPC system, reads what comes back, decides what to try next, and keeps going until it has an answer — for hours or days, with nobody in the loop.

One campaign per question

Each investigation is a campaign with its own goal, task definition and records. Campaigns are independent and run side by side, sharing the framework and the system definitions.

Visible and steerable

Any number of campaigns, all watchable from one Slack channel, and redirectable while they run. Nobody stands up new infrastructure per question.

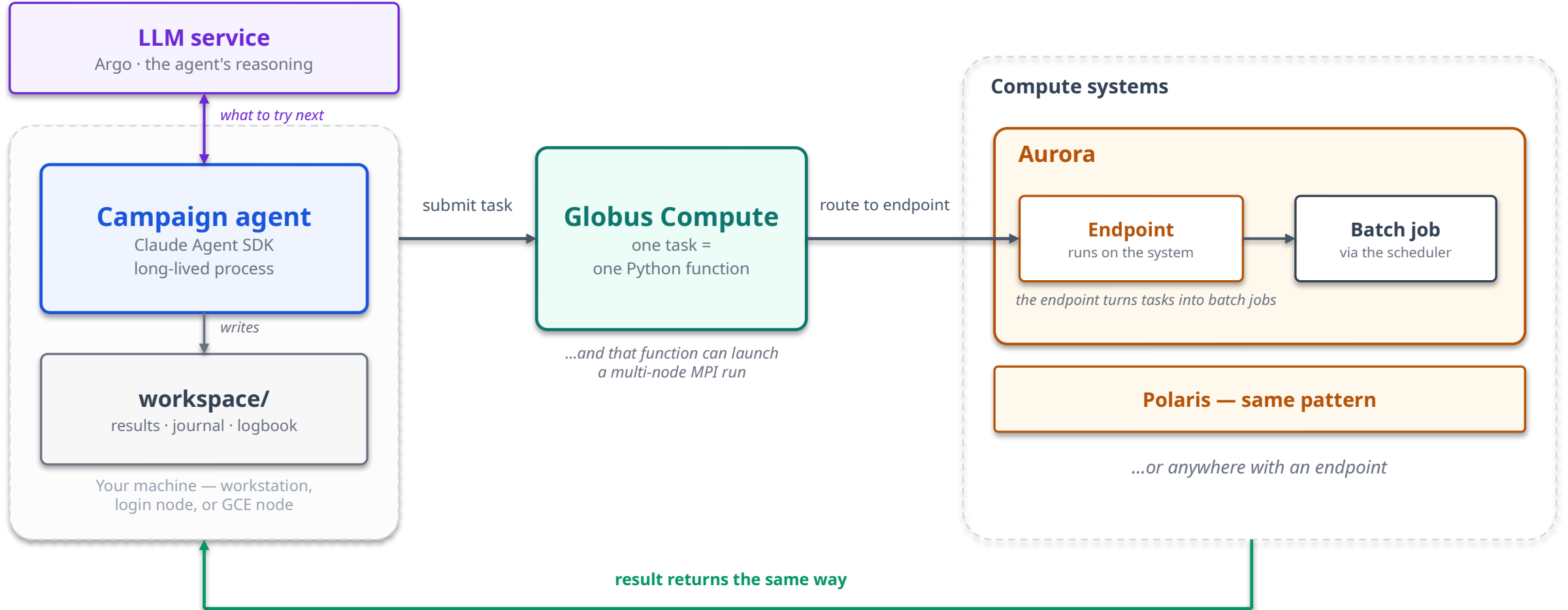
github.com/shuds13/AgentLab

built on the same machinery as github.com/shuds13/cas-framework, which coordinates many agents on a single search campaign

Runs anywhere the Claude Agent SDK and a Globus Compute endpoint can be reached — the agent does not need to be on the HPC system.

How a campaign runs

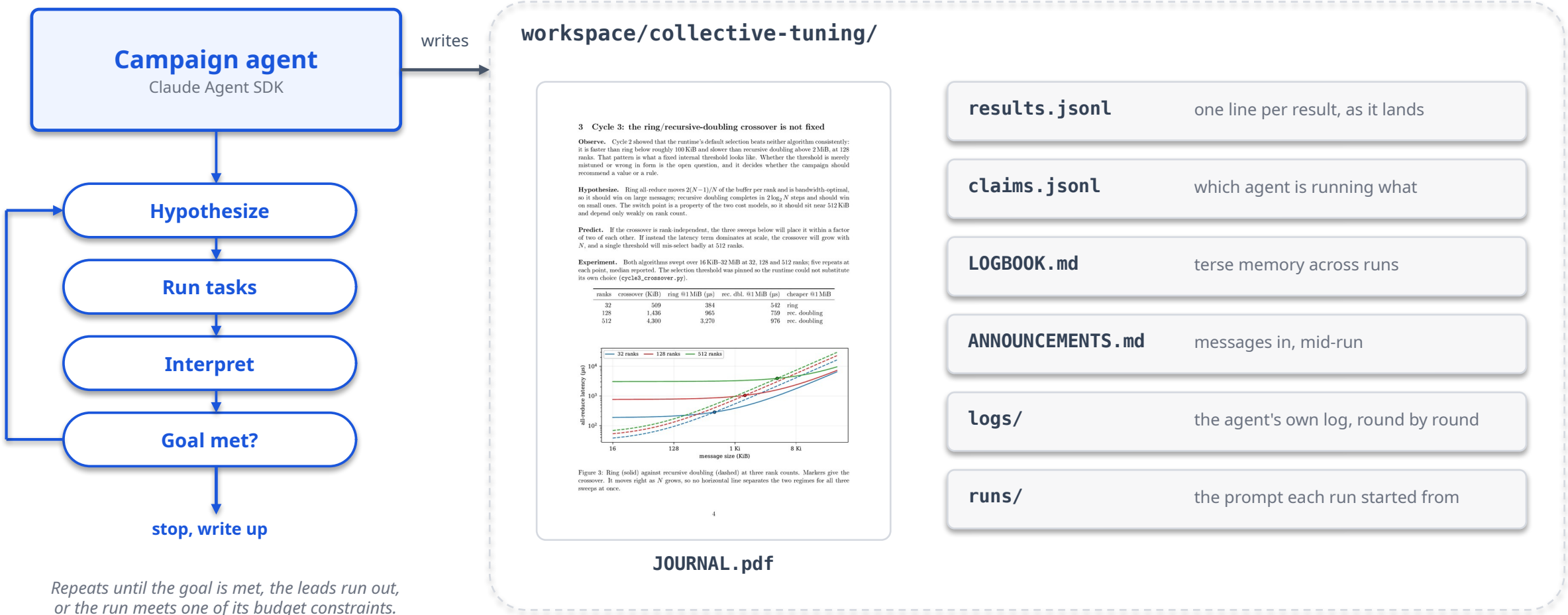
You give an agent a goal and a system. It runs the investigation itself.



The loop is the point: submit, read, decide, submit again — for hours or days, without a person in it.

The workspace

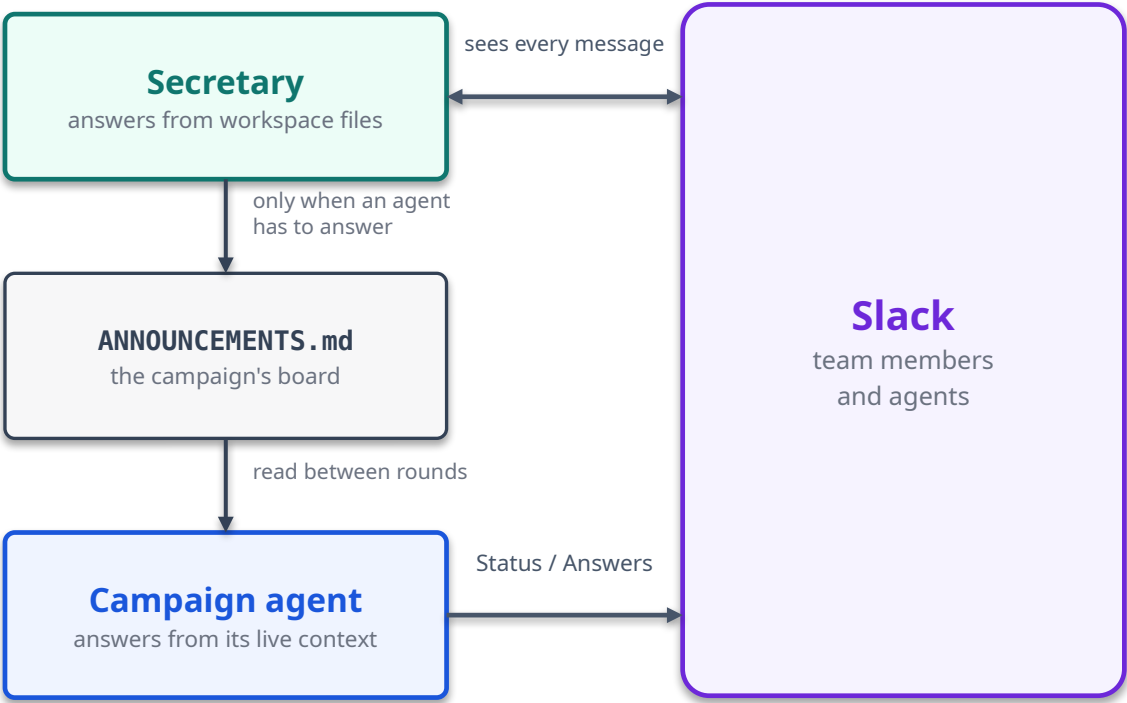
One directory per campaign. Everything the agent produces lands in it.



Several agents can share one workspace — `claims.jsonl` is how they avoid running the same work twice.

Human in the loop

The lab runs on its own. You can still watch it, question it, and redirect it — from Slack.



You can put an announcement in ANNOUNCEMENTS.md for the agents at any time.

The screenshot shows a Slack thread with four messages. The first is from **qaoo_agent** (APP) at 12:34 PM, stating that the **[polaris/explore-1234]** agent is exiting because the **globus-compute-endpoint 'qaoo-polaris'** is offline, and provides a command to start it. The second is from **qaoo_agent** (APP) at 1:12 PM, announcing that the **[polaris/explore-1312]** agent has started using **claude-opus-5[1m]**. The third is from **Stephen Hudson** at 1:14 PM, asking **@qaoo_agent** what prompt the polaris agent is using. The fourth is from **qaoo_agent** (APP) at 1:14 PM, responding that the **[secretary]** is running **polaris_explore_20260812_131237** with the standard user prompt.

an alert, a start notice, a question, and the secretary answering it

The secretary answers from the workspace files; the research agents answer from their live context.

Getting started

Clone the repository, start your agent in it, and ask.

1 Clone it

```
git clone  
https://github.com/shuds13/AgentLab
```

2 Start your agent in it

```
cd AgentLab && claude
```

3 Say

"Help me set up a campaign."

What you need

- Claude Agent SDK, and an LLM service it can reach
- Python 3.10+, and the claude CLI on PATH and authenticated
- Access to a compute system, and a project to charge

Where to run

- Preferably a shared persistent node, so the team reaches the same campaigns
- Inside tmux. A campaign runs for hours or days, and closing the terminal kills it with jobs still in flight

AGENTS.md in the repository root is the onboarding script — the agent reads it and walks you through the rest.